

COURSE PROJECT REPORT

Stroke Prediction Using Data Mining Techniques

Machine Learning-Based Stroke Risk Prediction

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Data Mining Course

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Abstract

Stroke is a major health condition in which early identification of individuals with higher stroke risk may support further medical investigation and preventive strategies. This project applies data mining and machine learning techniques to predict whether a patient belongs to the stroke or no-stroke class using demographic, medical, and lifestyle attributes. The study uses a Kaggle stroke prediction dataset containing 5,110 patient records. The workflow includes exploratory data analysis, missing-value treatment, categorical encoding, numerical scaling, train-test splitting, SMOTE-based class balancing, model comparison, feature importance analysis, and feature selection. Five classifiers were evaluated: Logistic Regression, Decision Tree, Random Forest, K-Nearest Neighbors, and XGBoost. After reducing the processed feature set from 21 to 10 selected features, the final experiment used Logistic Regression. On the unseen test set, the final model achieved a ROC-AUC of 0.842 and a Recall of 84%. The confusion matrix contained 42 true positives, 8 false negatives, 249 false positives, and 723 true negatives. The project also includes an interactive prediction interface that accepts patient information and returns a predicted class and estimated stroke probability. The results demonstrate a complete data mining workflow while also highlighting the limitations caused by class imbalance and false-positive predictions.

1. Introduction

Data mining provides methods for discovering useful patterns in large datasets and using those patterns for prediction and decision support. In healthcare-related applications, machine learning can be used to analyze patient information and identify statistical patterns associated with medical outcomes.

This project focuses on stroke prediction as a binary classification problem. The objective is not to diagnose patients, but to investigate whether routinely recorded demographic, medical, and lifestyle variables can be used to build a predictive model.

1.1 Problem Statement

The proposed problem is to predict whether a patient is likely to belong to the stroke class based on available patient characteristics. The target variable is binary: 0 represents No Stroke and 1 represents Stroke.

1.2 Project Objectives

- Understand the structure and characteristics of the stroke dataset.
- Explore relationships between patient attributes and stroke occurrence.
- Clean and preprocess the dataset for machine learning.
- Handle the strong class imbalance using SMOTE on the training data.
- Compare five machine learning classification algorithms.
- Use feature importance and feature selection to reduce the processed feature space.
- Evaluate the final model using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and a Confusion Matrix.
- Develop an interactive prediction interface for new patient inputs.

2. Dataset Description

The project uses the Stroke Prediction Dataset available through Kaggle. The dataset contains 5,110 patient records and demographic, medical, and lifestyle attributes.

Dataset source: Kaggle – Stroke Prediction Dataset

Kaggle URL: <https://www.kaggle.com/datasets/fedesoriano/stroke-prediction-dataset>

Feature	Type	Description
gender	Categorical	Patient gender
age	Numerical	Patient age
hypertension	Binary	Hypertension indicator
heart_disease	Binary	Heart disease indicator
ever_married	Categorical	Whether the patient has ever married
work_type	Categorical	Employment category
Residence_type	Categorical	Urban or Rural residence
avg_glucose_level	Numerical	Average glucose level
bmi	Numerical	Body Mass Index
smoking_status	Categorical	Smoking category
stroke	Target	0 = No Stroke, 1 = Stroke

3. Exploratory Data Analysis

Exploratory Data Analysis (EDA) was performed before modeling to understand the data distribution and examine relationships between selected variables and stroke status.

3.1 Target Distribution

The target variable is highly imbalanced. There are 4,861 No Stroke records (95.13%) and 249 Stroke records (4.87%). This imbalance is important because a classifier could obtain high accuracy by favoring the majority class while performing poorly on stroke detection.

3.2 Main EDA Findings

- The stroke group generally shows a higher age distribution than the no-stroke group.
- Average glucose level shows a wider and generally higher distribution among stroke cases.
- BMI distributions overlap considerably between stroke and no-stroke groups.
- The observed stroke rate is higher among patients with hypertension.
- The observed stroke rate is higher among patients with heart disease.
- Stroke rates vary across smoking-status categories.
- The displayed correlation analysis shows age as the strongest correlation with stroke among the listed numerical/binary variables (0.25), followed by hypertension, heart disease, and average glucose level (0.13 each).

4. Data Preprocessing

4.1 Missing Values

The BMI feature contained 201 missing values, corresponding to 3.93% of the 5,110 records. The missing BMI values were replaced using median imputation. The calculated BMI median was 28.1. After imputation, no missing BMI values remained.

4.2 Duplicate Check

No duplicate rows were found in the dataset.

4.3 Removing the ID

The id column was removed because it is an identifier and does not represent a meaningful patient characteristic for prediction.

4.4 Categorical Encoding

Categorical variables were transformed using One-Hot Encoding so that machine learning algorithms could process them numerically.

4.5 Numerical Scaling

The numerical features age, avg_glucose_level, and bmi were standardized using StandardScaler.

4.6 Train-Test Split

The data was divided into training and testing subsets using an 80/20 stratified split. The test set was kept separate from the training process and used for final evaluation.

5. Handling Class Imbalance with SMOTE

Because stroke cases represent only 4.87% of the original dataset, SMOTE (Synthetic Minority Over-sampling Technique) was applied to the training data. SMOTE creates synthetic minority-class examples to reduce the imbalance during model training.

SMOTE was applied only after the train-test split and only to the training data. This prevents synthetic information from influencing the unseen test set.

After SMOTE, the training data contained 7,778 samples: 3,889 No Stroke and 3,889 Stroke.

6. Machine Learning Algorithms

Five classification algorithms were selected for comparison:

1. Logistic Regression – used as a strong and interpretable baseline for binary classification.
2. Decision Tree – provides rule-based classification and can model nonlinear relationships.
3. Random Forest – an ensemble of decision trees that can capture nonlinear patterns and provide feature importance.

4. K-Nearest Neighbors (KNN) – predicts a class based on similarity to nearby observations.
5. XGBoost – a gradient-boosting ensemble method that can model complex nonlinear relationships.

7. Feature Importance and Feature Selection

After one-hot encoding, the model input contained 21 processed features. Feature selection was then used to identify a smaller set of informative variables. SelectKBest with mutual_info_classif was used for this experiment.

The feature space was reduced from 21 processed features to 10 selected features.

Selected features:

- age
- bmi
- ever_married_No
- ever_married_Yes
- work_type_Private
- smoking_status_Unknown
- smoking_status_formerly smoked
- smoking_status_never smoked
- hypertension
- heart_disease

Random Forest feature-importance analysis also showed age as the most important displayed feature (importance ≈ 0.3186), followed by average glucose level (≈ 0.1148), BMI (≈ 0.1082), and hypertension (≈ 0.0647). Feature importance is model-specific and should not be interpreted as medical causation.

8. Model Evaluation

The models were evaluated using Accuracy, Precision, Recall, F1-Score, ROC-AUC, and Confusion Matrix. Because the target is highly imbalanced, particular attention was given to Recall and ROC-AUC.

8.1 Model Comparison

The project compared the five algorithms before and after feature selection. The observed F1-score comparison showed different responses to reducing the feature space. The final experiment selected Logistic Regression with the 10 selected features.

8.2 Final Model Performance

Metric	Test Result
Accuracy	74.85%
Precision	14.43%
Recall	84.00%
F1-Score	24.49%

ROC-AUC	0.842
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9. Final Confusion Matrix Analysis

The final Logistic Regression model with 10 selected features produced the following confusion matrix on the unseen test set:

Actual / Predicted	No Stroke	Stroke
No Stroke	723	249
Stroke	8	42

- True Negative (TN) = 723
- False Positive (FP) = 249
- False Negative (FN) = 8
- True Positive (TP) = 42

The model correctly identified 42 of the 50 actual stroke cases in the test set, giving a Recall of 84%. However, it also generated 249 false positives. This trade-off is important when interpreting the model and shows why multiple evaluation metrics are necessary.

10. ROC Curve Analysis

The final ROC curve produced an AUC of 0.842. ROC-AUC summarizes the model's ability to distinguish between stroke and no-stroke cases across different probability thresholds. A value of 0.842 indicates useful discrimination on this test set, while not implying clinical validity.

11. Interactive Prediction System

An interactive interface was developed using Python and ipywidgets. The interface allows a user to enter patient information and request a stroke prediction.

- Gender
- Age
- Hypertension
- Heart Disease
- Marital Status
- Work Type
- Residence Type
- Average Glucose Level
- BMI
- Smoking Status

The system processes the entered values using the same preprocessing and selected-feature pipeline and returns a predicted class together with an estimated stroke probability.

The prediction is a machine-learning estimate based on patterns learned from the dataset. It is not a medical diagnosis and should not be used for clinical decision-making.

12. Discussion

The project demonstrates how a complete data mining pipeline can be constructed for an imbalanced binary classification problem. Exploratory analysis identified meaningful differences in the distributions of age, glucose, hypertension, heart disease, and smoking status between the observed groups.

The use of SMOTE addressed the imbalance during model training, while the original test set was preserved for evaluation. Feature selection reduced the processed feature space from 21 to 10 features, making the final experiment more compact.

The final model achieved high recall relative to its other metrics and correctly detected 42 of the 50 stroke cases in the test set. However, the precision was low and the model generated many false positives. For a real medical system, the threshold, costs of false positives and false negatives, external validation, and clinical requirements would need to be studied carefully.

13. Limitations

6. The target variable is strongly imbalanced, with only 4.87% stroke cases.
7. The final model produced a relatively high number of false positives.
8. The dataset and evaluation setup are not sufficient to establish clinical effectiveness.
9. The model should be tested on independent external datasets before any real-world application.
10. Feature importance and statistical associations do not prove causality.
11. Performance can depend on preprocessing, feature selection, class-balancing, model parameters, and threshold choice.

14. Future Work

12. Perform hyperparameter optimization using Grid Search or Randomized Search.
13. Use cross-validation for more robust performance estimates.
14. Compare additional ensemble and machine learning algorithms.
15. Evaluate alternative class-imbalance techniques.
16. Investigate additional feature engineering.
17. Test the final pipeline on independent external datasets.
18. Add explainable AI techniques to provide explanations for individual predictions.
19. Analyze false-positive and false-negative cases in more detail.
20. Develop a web-based deployment of the prediction interface.
21. Perform appropriate clinical validation before any real-world use.

15. Conclusion

This project implemented a complete data mining workflow for stroke prediction. The workflow included dataset understanding, exploratory data analysis, missing-value treatment, duplicate checking, identifier removal, categorical encoding, numerical scaling, stratified train-test splitting, SMOTE-based balancing, machine learning model comparison, feature importance analysis, feature selection, final evaluation, and an interactive prediction interface.

The final experiment used Logistic Regression with 10 selected features. On the unseen test set, the model achieved Accuracy of 74.85%, Precision of 14.43%, Recall of 84.00%, F1-Score of approximately 24.49%, and ROC-AUC of 0.842. The model detected 42 of 50 stroke cases but produced 249 false positives.

Overall, the project demonstrates the practical application of data mining techniques to a healthcare-related prediction problem while emphasizing the importance of class-imbalance handling and appropriate evaluation metrics.

17. Project Requirements Coverage

Course Requirement	Project Implementation
1. Propose a problem	Stroke prediction as a binary classification problem.
2. Find a suitable dataset	Kaggle Stroke Prediction Dataset; 5,110 records and documented features.
3. Preprocess the dataset	Missing-value treatment, duplicate check, ID removal, encoding, scaling, train-test split, SMOTE, feature selection.
4. Select techniques and algorithms	Logistic Regression, Decision Tree, Random Forest, KNN, and XGBoost.
5. Select a tool	Python in Google Colab with machine-learning libraries.
6. Document results and findings	Metrics, confusion matrix, ROC curve, discussion, limitations, and future work.
7. Document every step	Notebook, report, presentation, and README.
8. Upload materials to GitHub	Repository structure prepared for notebook, README, dataset, images, and requirements.